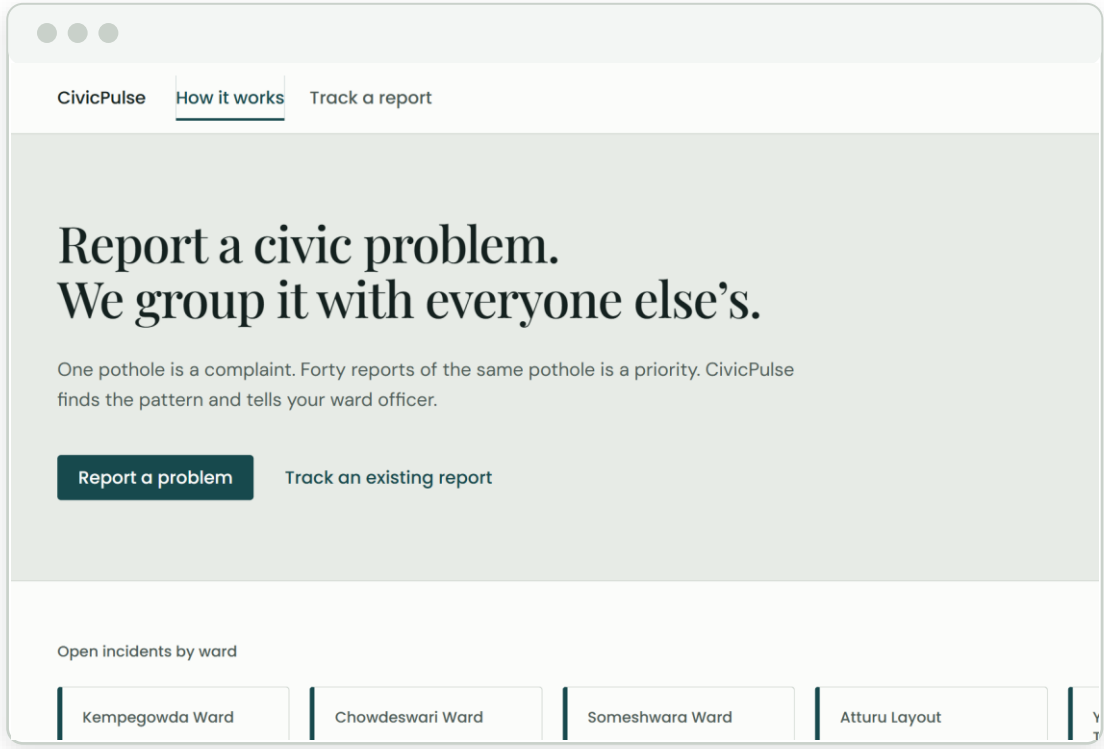
TEAM PROJECT · BENGALURU · 2026

CivicPulse

From scattered complaints to prioritised incidents

Citizens report problems in their own words. CivicPulse groups the reports that describe the same fault and tells the officer which to fix first, and why.



Suryansh · Abhishek · Devansh
Manya · Ria · Alaafiya

One pothole, reported five different ways

Huge pothole outside the 4th Block bus stop, very dangerous at night	Formal
road broken near bus stand. bike skidded	Terse
Third time complaining!! Nobody fixes this road	Angry
Bus stop ke paas bada gaddha hai, please repair	Code-mixed
Deep hole on the main road, Jayanagar 4th Block	Different words

Five tickets in the queue. *One road to fix.*

Duplicates bury the queue

Officers triage the same fault again and again while new problems wait.

Category grouping is too blunt

A blocked sewer and a missing drain cover share a category, but not a repair.

No reason behind "urgent"

Without a stated rule, ranking is a guess no one can check.

Report once. Group by meaning. Fix what matters first

01 Report

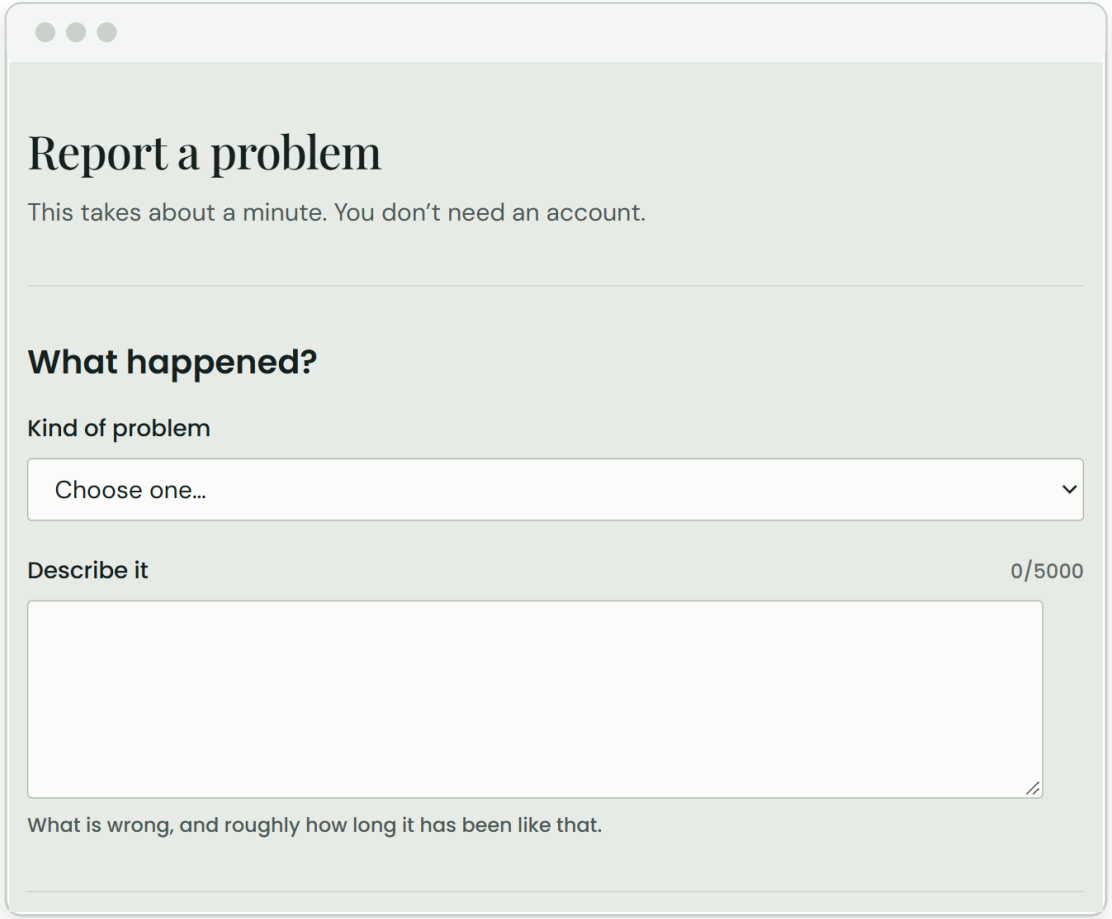
A citizen picks a category, drops a pin and describes the problem. No account is needed. The ward is found from 243 real BBMP ward boundaries.

02 Group

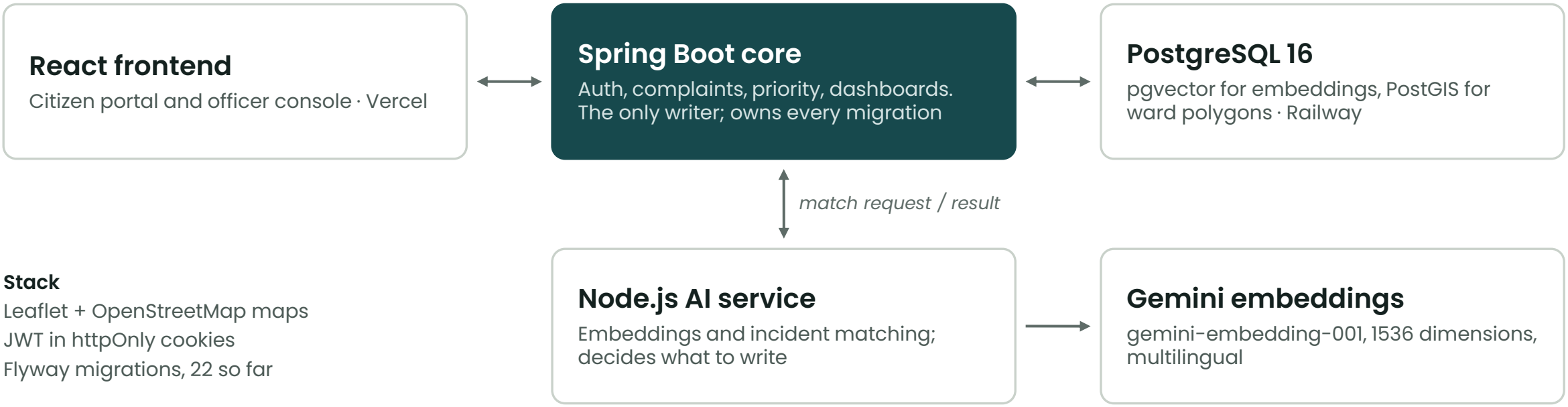
The text is turned into an embedding. Reports that mean the same thing, in the same ward and category and within 500 m, join one incident.

03 Prioritise and act

Each incident gets a score with written reasons. Officers work a ranked queue, a map, hotspots and a suggested next step.



Three services, one writer



Stack
Leaflet + OpenStreetMap maps
JWT in httpOnly cookies
Flyway migrations, 22 so far

Built so a citizen never waits

Save first
The complaint is stored and answered at once. Matching runs afterwards.

Fall back
If the AI service is down, a circuit breaker groups by ward and category, and marks it.

Repair
A sweep re-runs those matches when the AI service recovers.

Grouped by meaning, checked by distance

Same ward and category

Hard filter. Meaning never crosses a category.

Similar meaning, at least 0.84

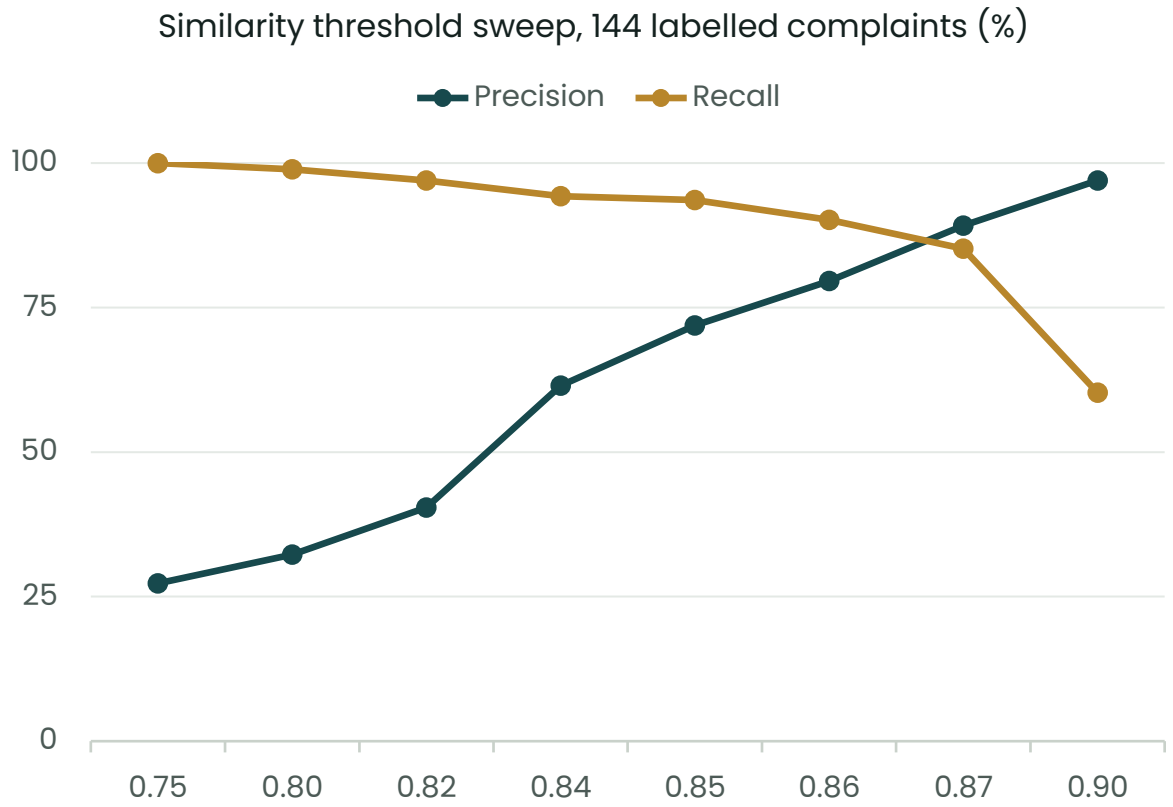
Cosine similarity to the closest report already in the incident.

Within 500 m

Same words a street away is a different fault.

55% → **95%**

of real grouping decisions correct: at the old 0.75 threshold, then at 0.84 with the 500 m check



At 0.75 every pool merged into one incident. **0.84 keeps recall above 94% and lifts precision to 62%.**

A score that shows its working



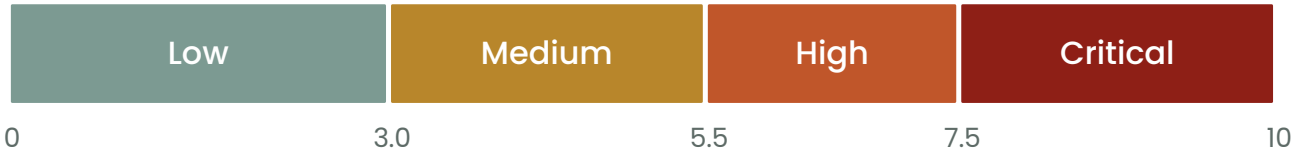
Volume
How many reports, full at 25

Growth
New reports in 24 h, full at 5

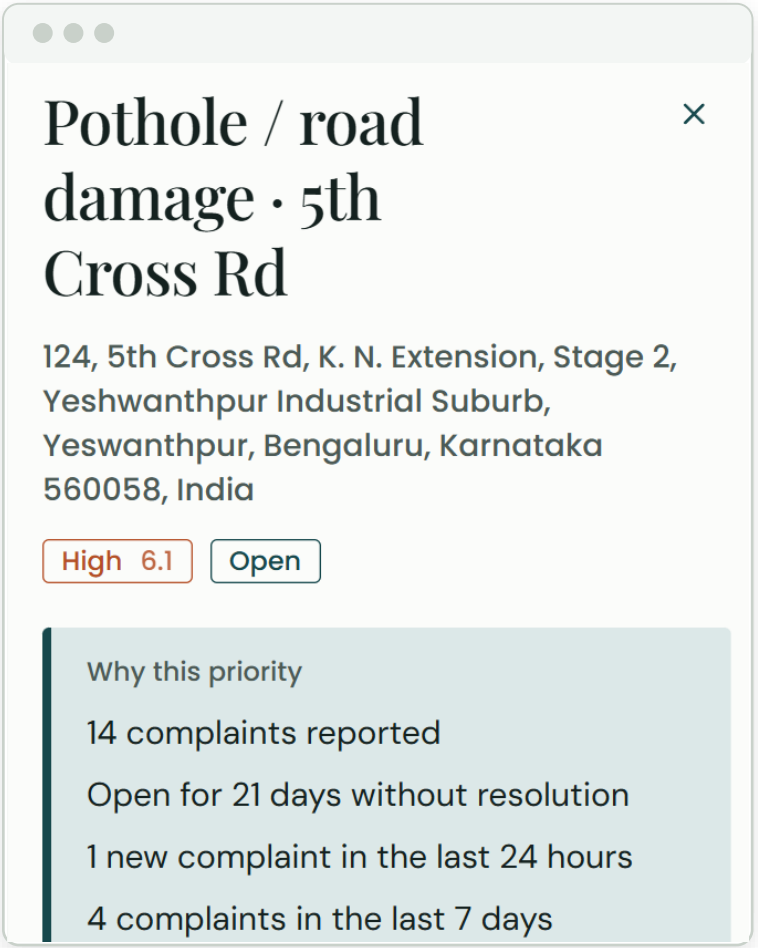
Age
Days since first report, full at 30

Spread
Distance between reports, full at 1 km

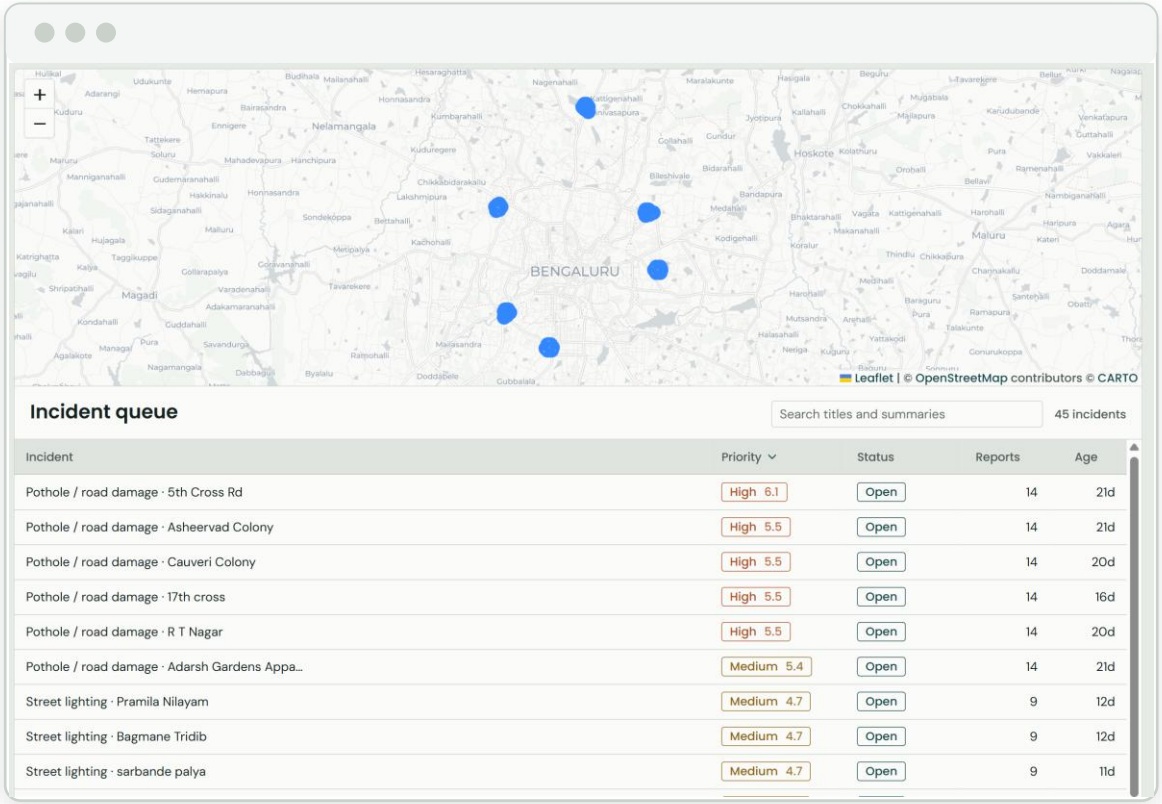
Score 0–10, banded



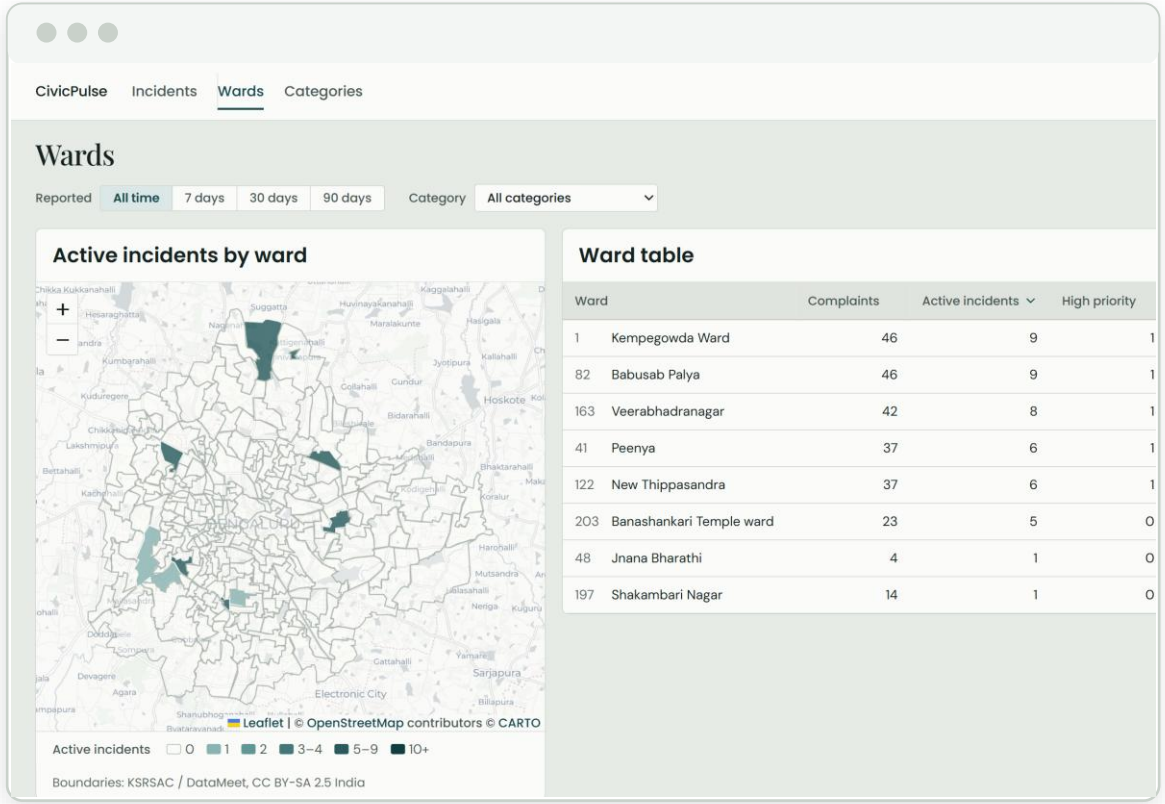
Every score is stored with its reasons, in plain words: "6 complaints reported", "2 new complaints in the last 24 hours", "Open for 12 days".



Everything an officer needs on one screen



Incident queue
249 demo reports grouped into 45 incidents, ranked by priority on a live map. Each opens with its reports and reasons.



Ward explorer
Active incidents across 243 BBMP wards as a map, with a sortable table of complaints, incidents and the main category.

We measured it instead of assuming it

95%

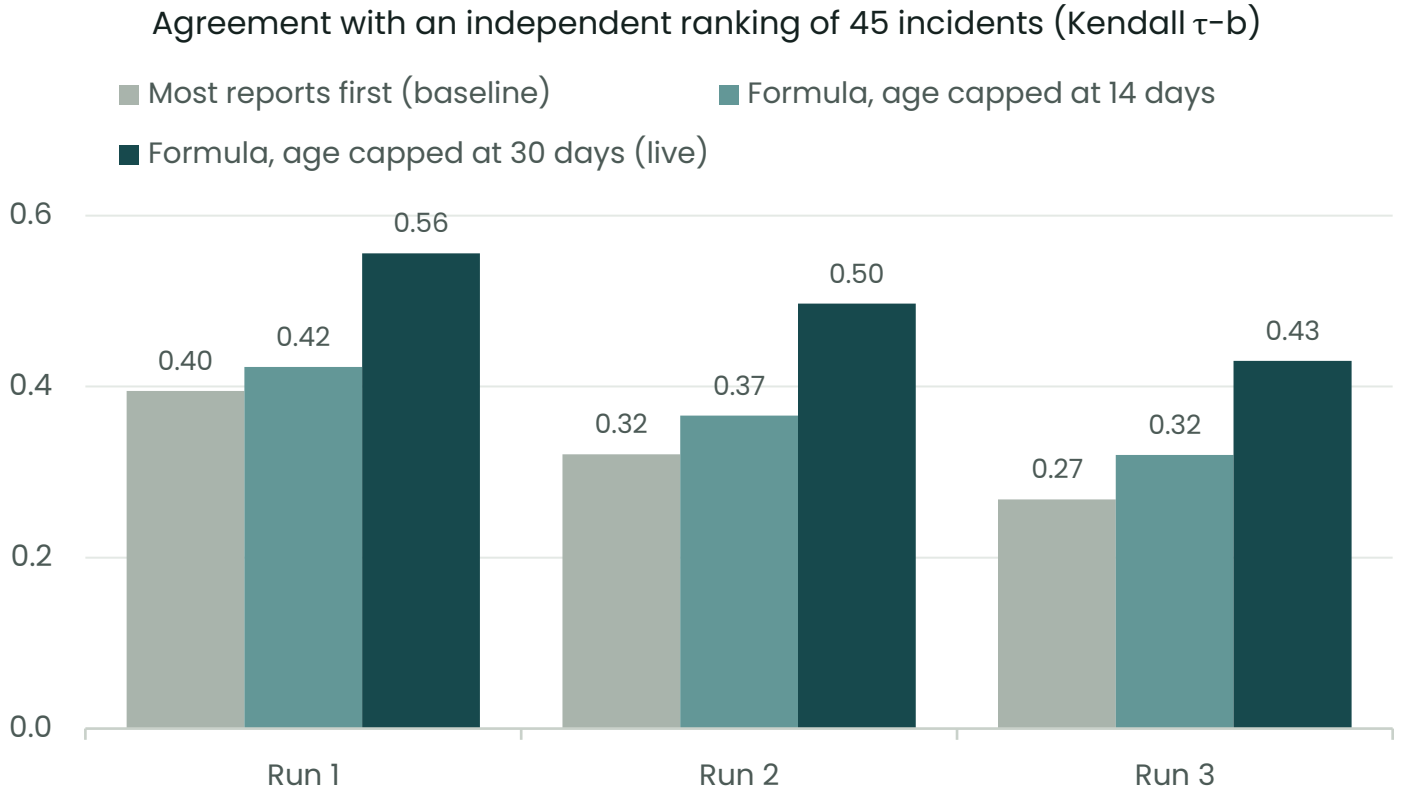
of real grouping decisions correct (was 55%)

3 of 3

ranking runs where the formula beat the "most reports first" baseline

243

real BBMP ward boundaries, matched with PostGIS



Ranker: Gemini, briefed as a ward engineer and never shown the formula. A human officer ranking is the next check.

Live today, and what comes next

Complaint intake
Anonymous reports, ward lookup, reference numbers

Semantic matching
Embeddings, 0.84 threshold, 500 m check

Explainable priority
Four terms, stored reasons, refreshed every 5 min

Officer console
Queue, map, incident detail, status changes

Hotspots and attention
Clusters, worsening and spiking signals

Decision support
Suggested next step, accept or override, logged

Next

Officer Copilot
Ask "what is urgent in Ward 48?" and get an answer that cites its incidents. Backend live; panel in progress.

Research paper
Matching and priority evaluation, written up.

Photo upload
Contract approved, not built.

Life-safety flag
Open drains, exposed wiring: escalate regardless of score.

Thank you

Questions welcome

Try it live
civic-pulse-gamma-ivory.vercel.app

Source
github.com/Proj-CivicPulse/CivicPulse

